



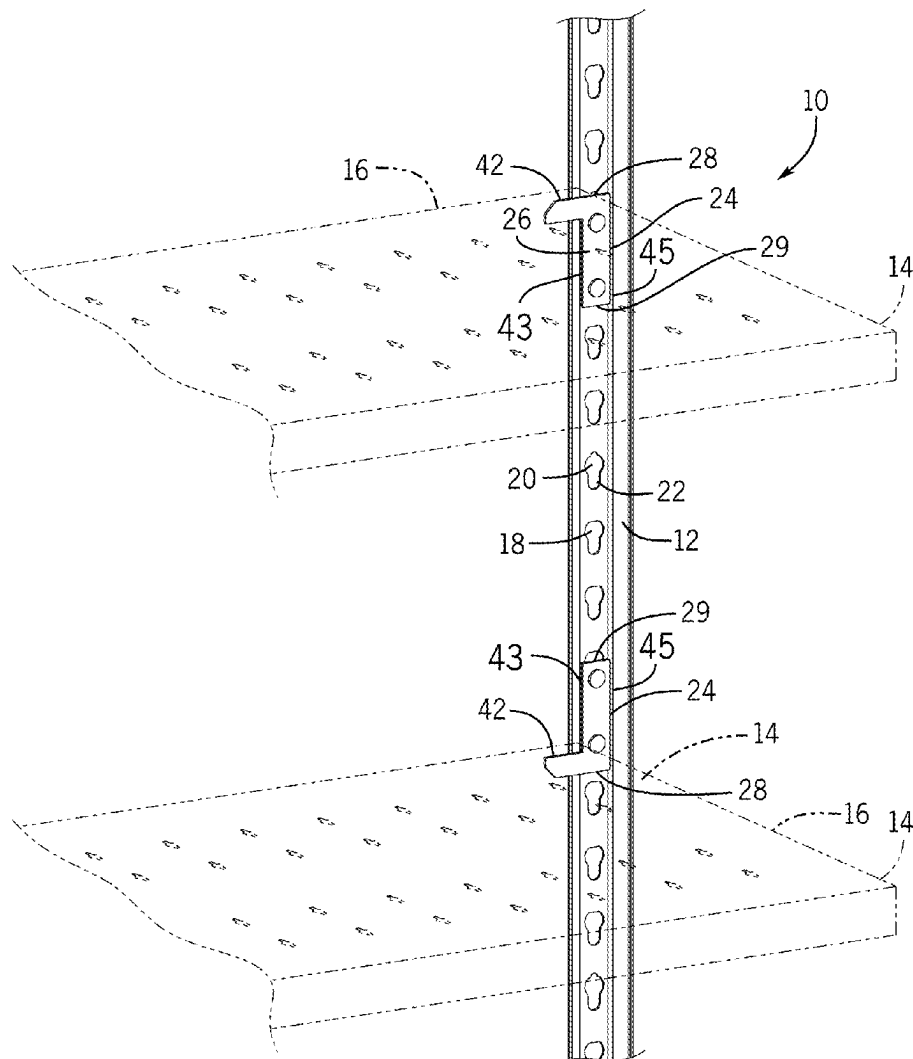
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(19) **United States**(12) **Patent Application Publication**
Wray et al.(10) **Pub. No.: US 2025/0275629 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **UNIVERSAL SHELF CLIP FOR STORAGE SYSTEM****Publication Classification**(51) **Int. Cl.**
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(52) **U.S. Cl.**
CPC *A47B 57/487* (2013.01)(71) Applicant: **SPACESAVER CORPORATION**, Fort Atkinson, WI (US)(72) Inventors: **Jessi L. Wray**, Janesville, WI (US);
Theodore E. Jach, Cambridge, WI (US); **Kenneth R. Durfee, JR.**, Janesville, WI (US); **Sean Krause**, Fort Atkinson, WI (US)(73) Assignee: **SPACESAVER CORPORATION**, Fort Atkinson, WI (US)(21) Appl. No.: **18/916,308**(22) Filed: **Oct. 15, 2024****Related U.S. Application Data**

(60) Provisional application No. 63/559,454, filed on Feb. 29, 2024.

(57) **ABSTRACT**

A storage system with reversible shelf clips that can be oriented in at least two different ways to achieve variable shelf spacing. The shelf clips have a support arm that extends laterally from a clip plate and a pair of locating pins that can be slidably attached to the clip plate and inserted into the cutouts of the vertical posts. The shelf clip is asymmetrical such that the orientation of the shelf clip can be changed such that the support arm can be positioned above or below the center line of the clip plate, depending on the desired shelf height, and the locating pins can protrude from either face surface of the clip plate, allowing the shelf clips to be used in multiple orientations. The shelf clips are universal and eliminate the need for multiple parts or handed components to set the shelf spacing.



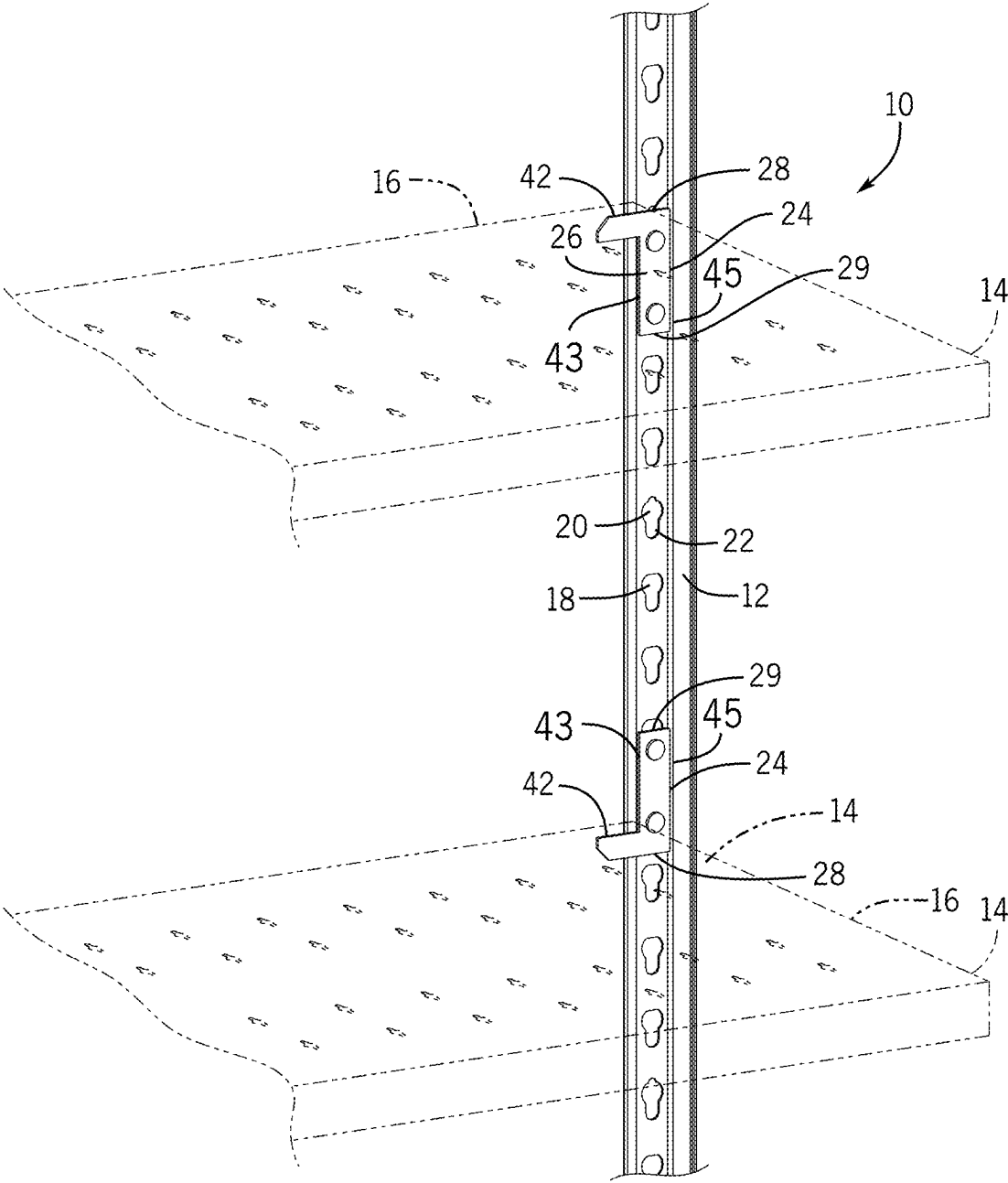


FIG. 1

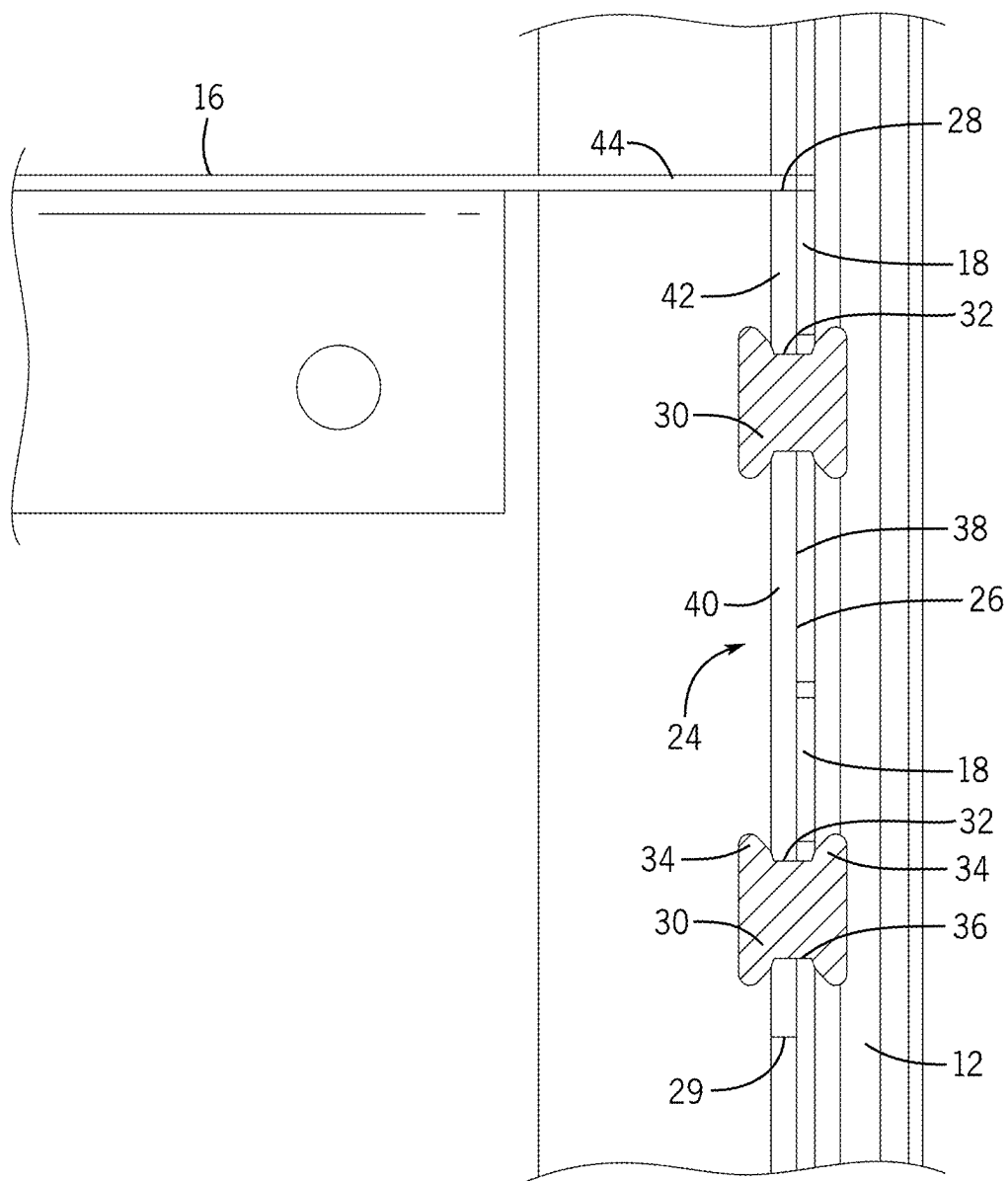


FIG. 2

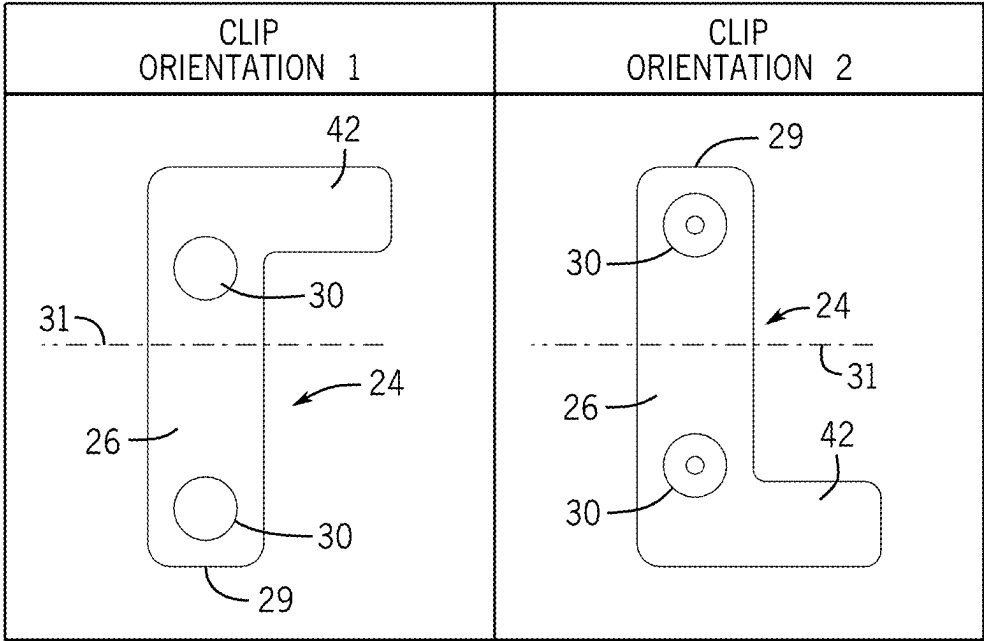


FIG. 3

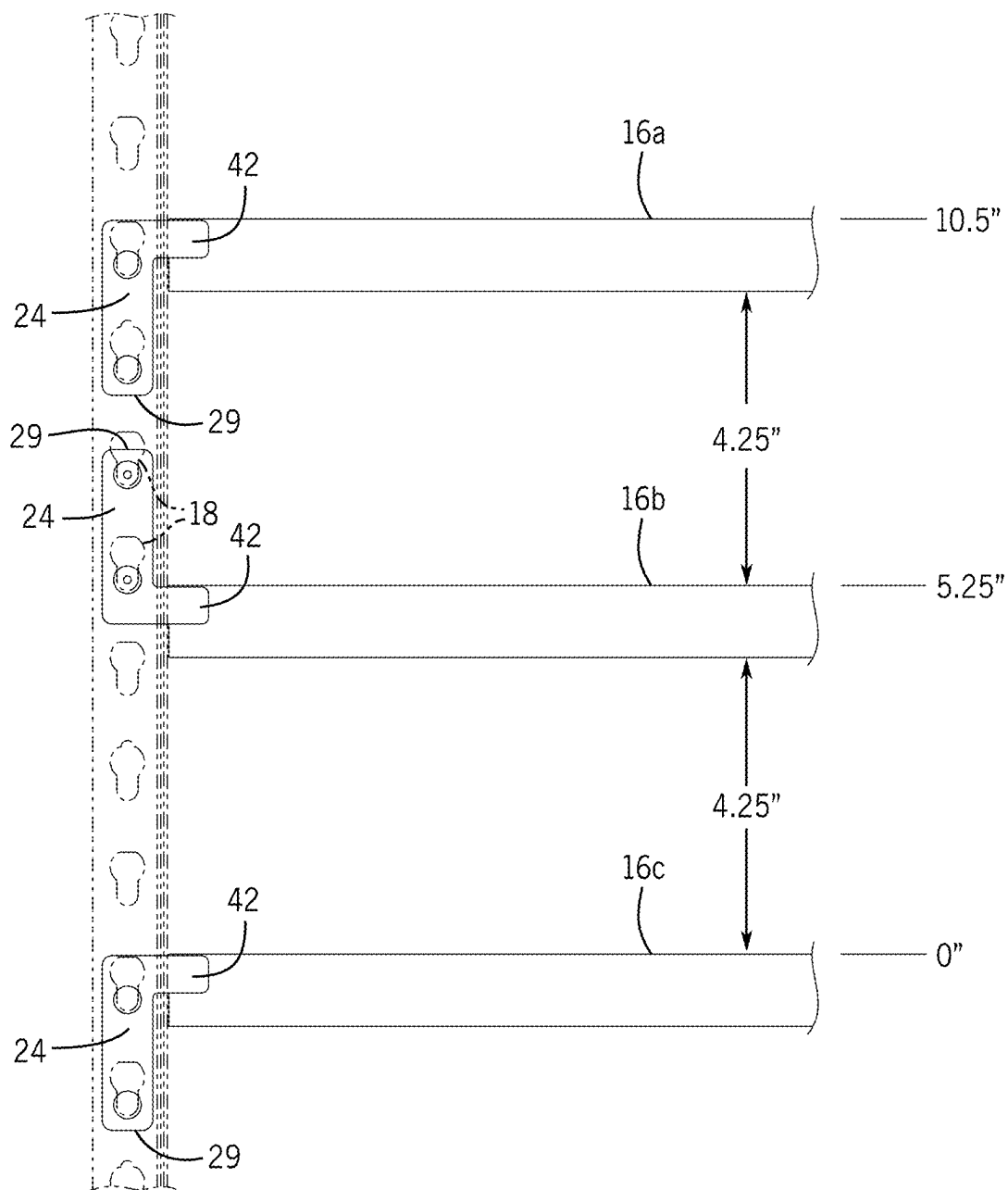


FIG. 4

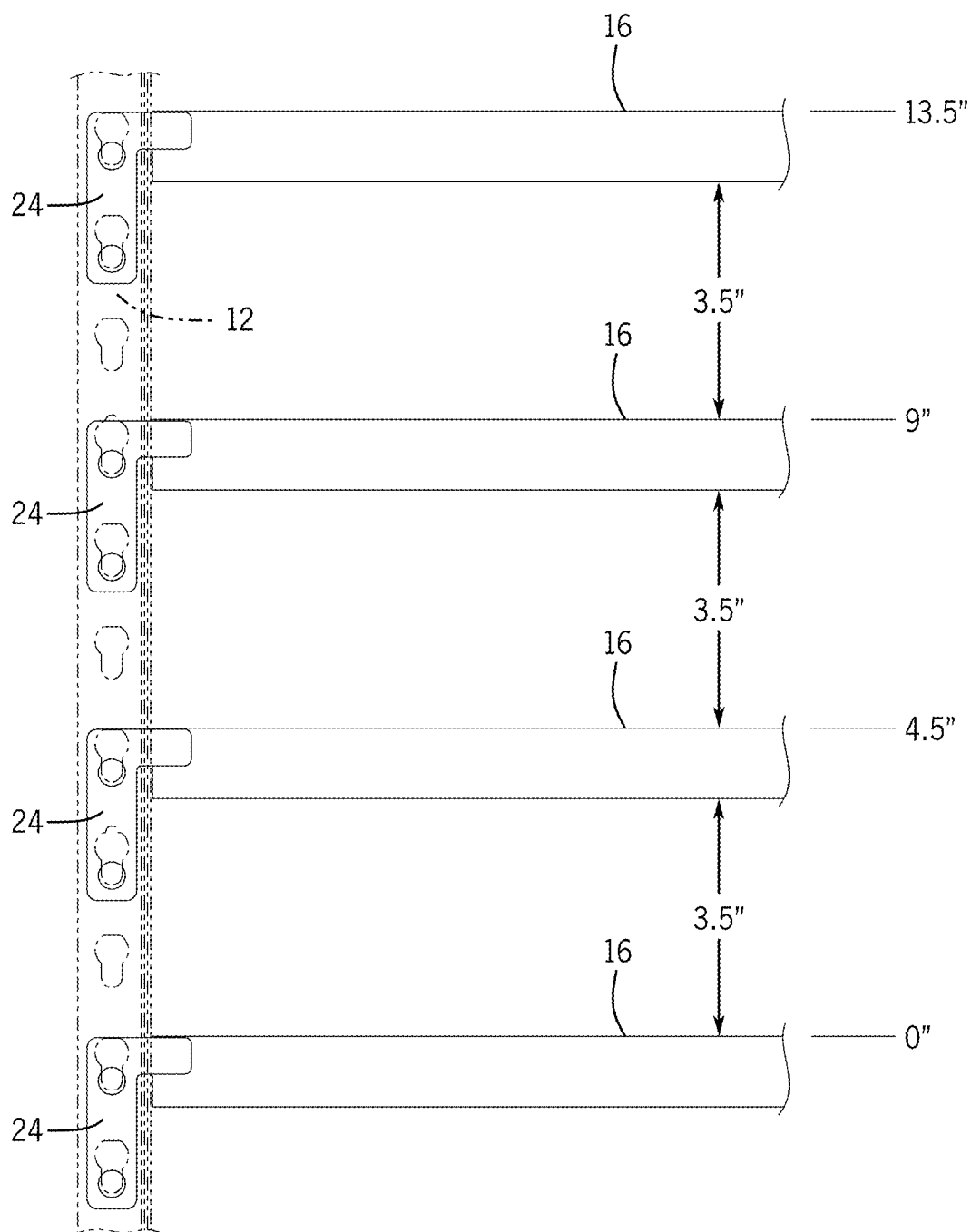
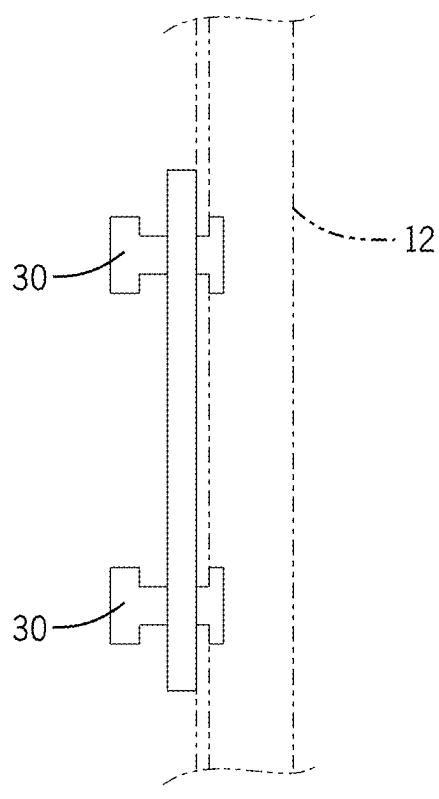
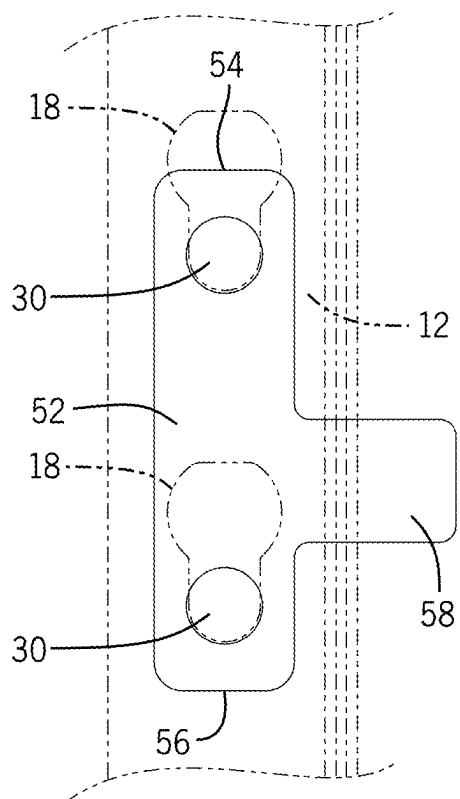
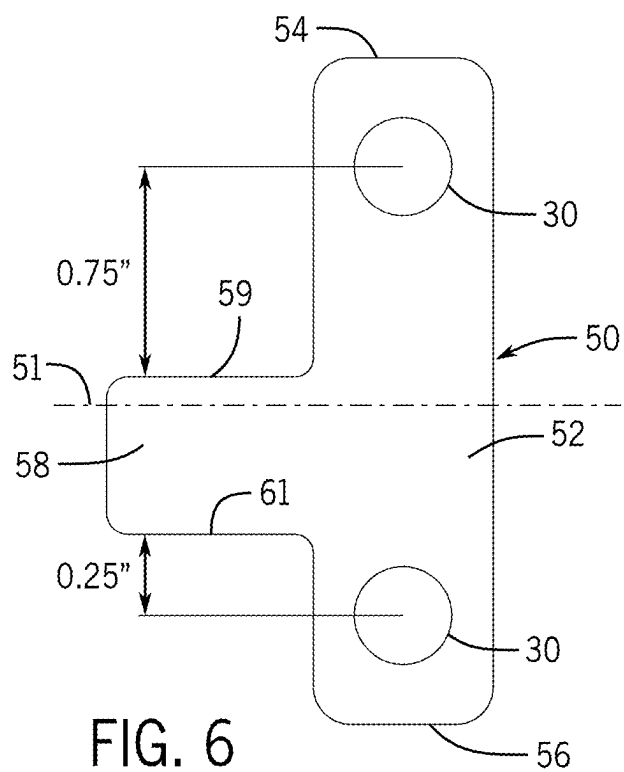


FIG. 5



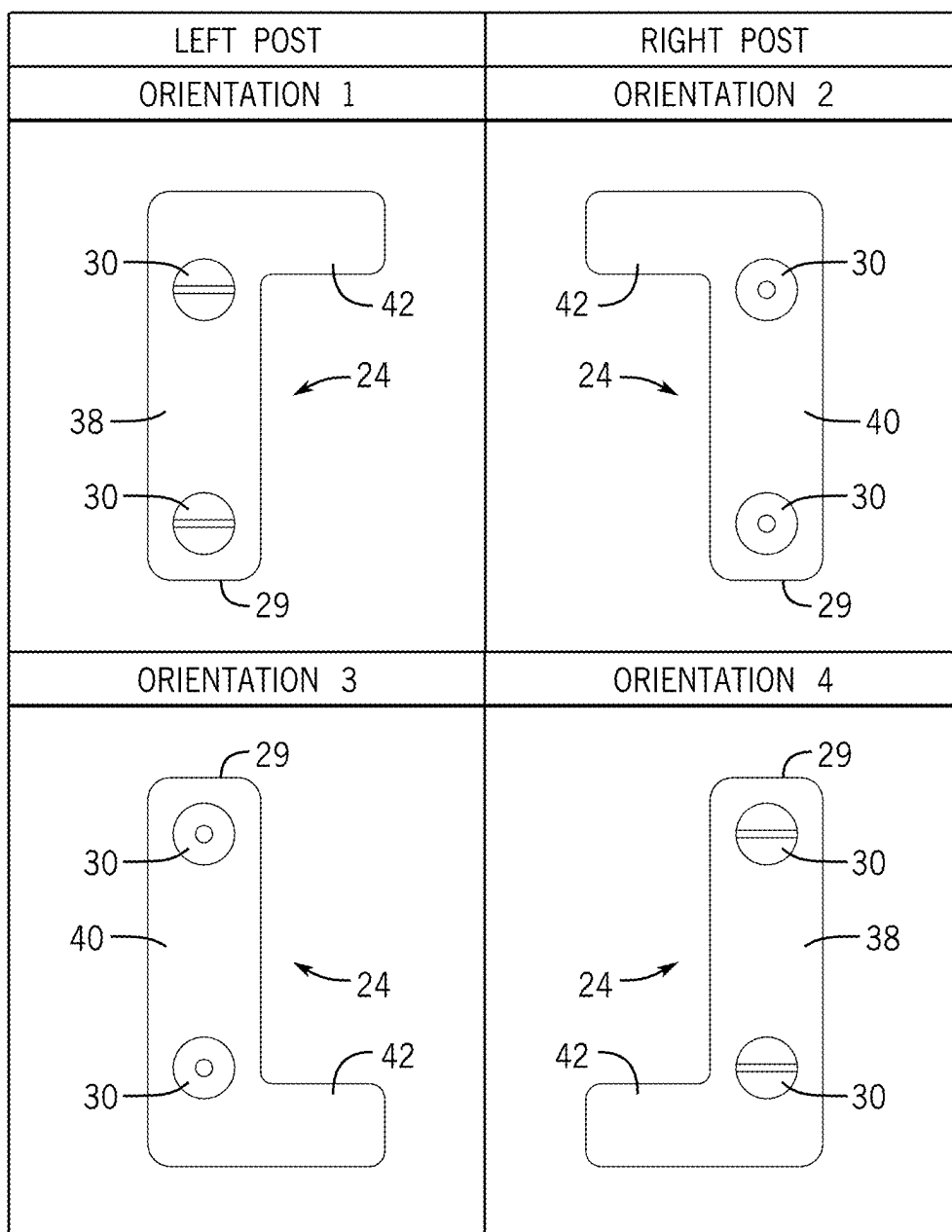


FIG. 9

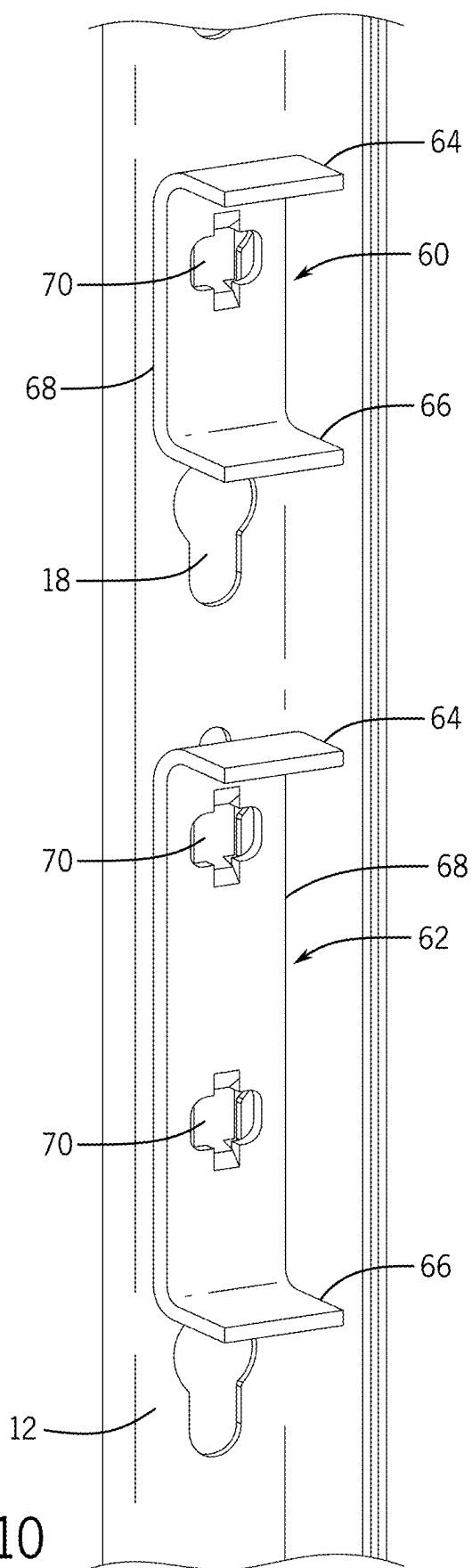


FIG. 10

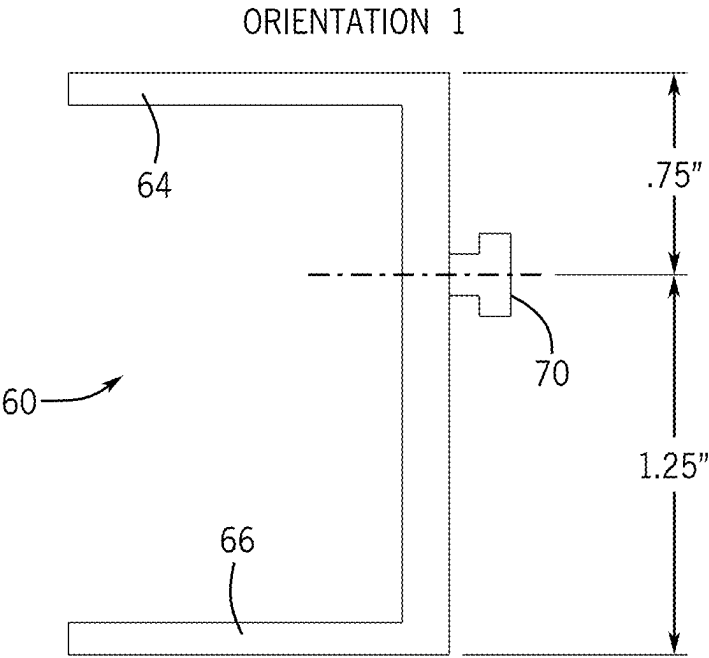


FIG. 11

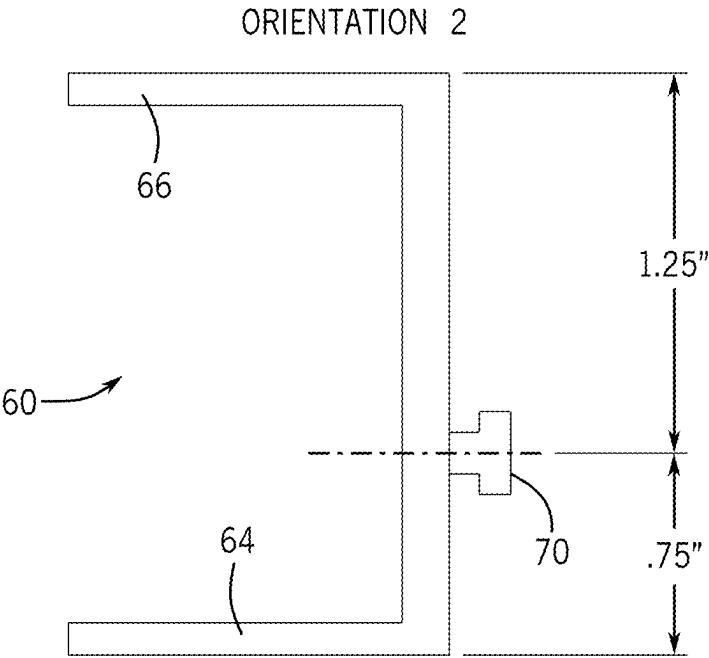


FIG. 12

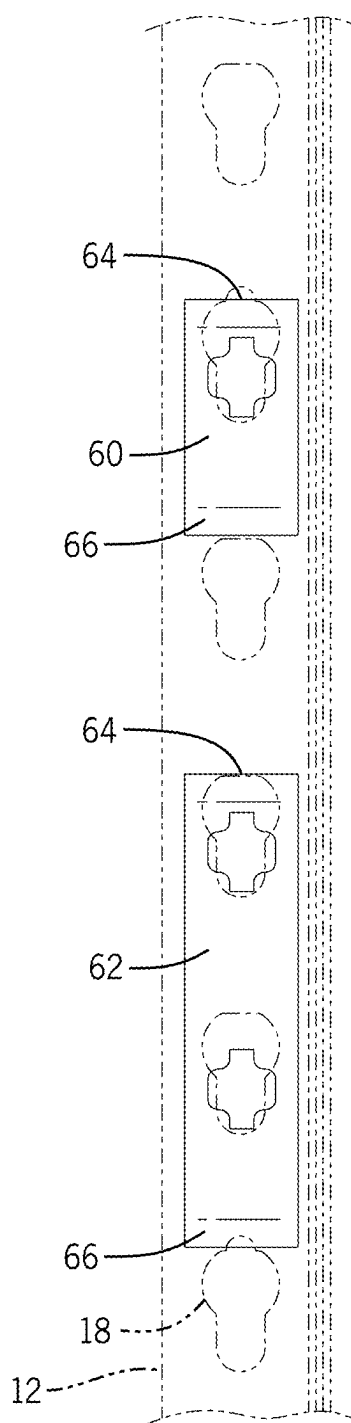


FIG. 13

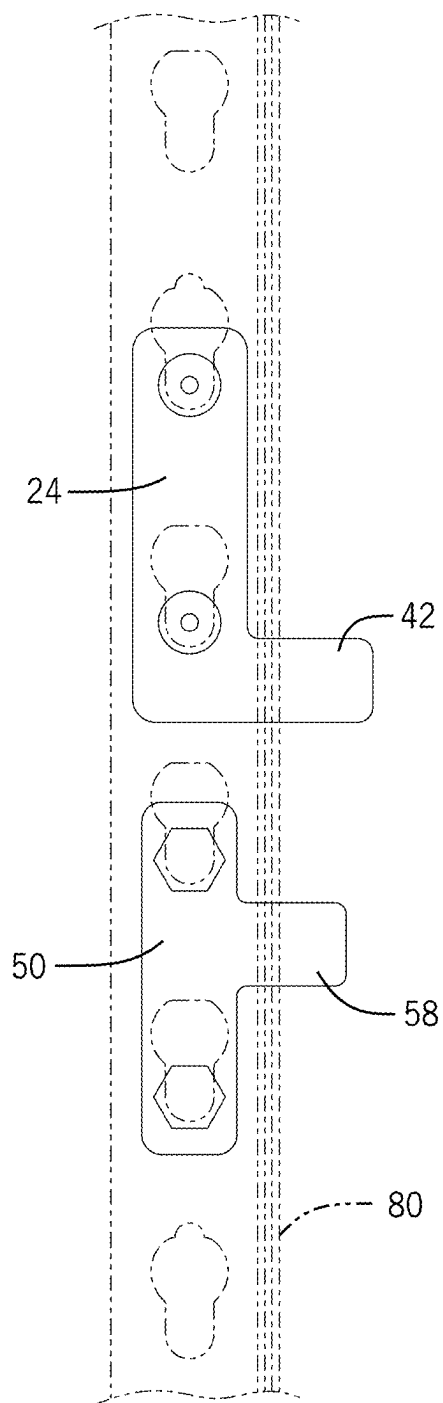


FIG. 14

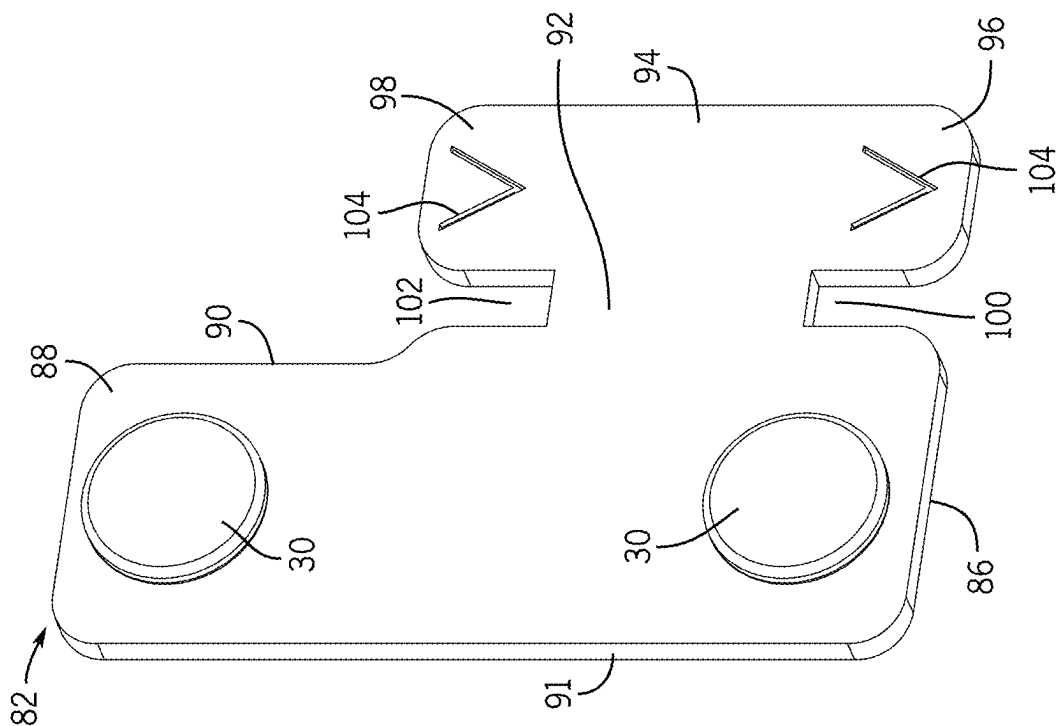


FIG. 15

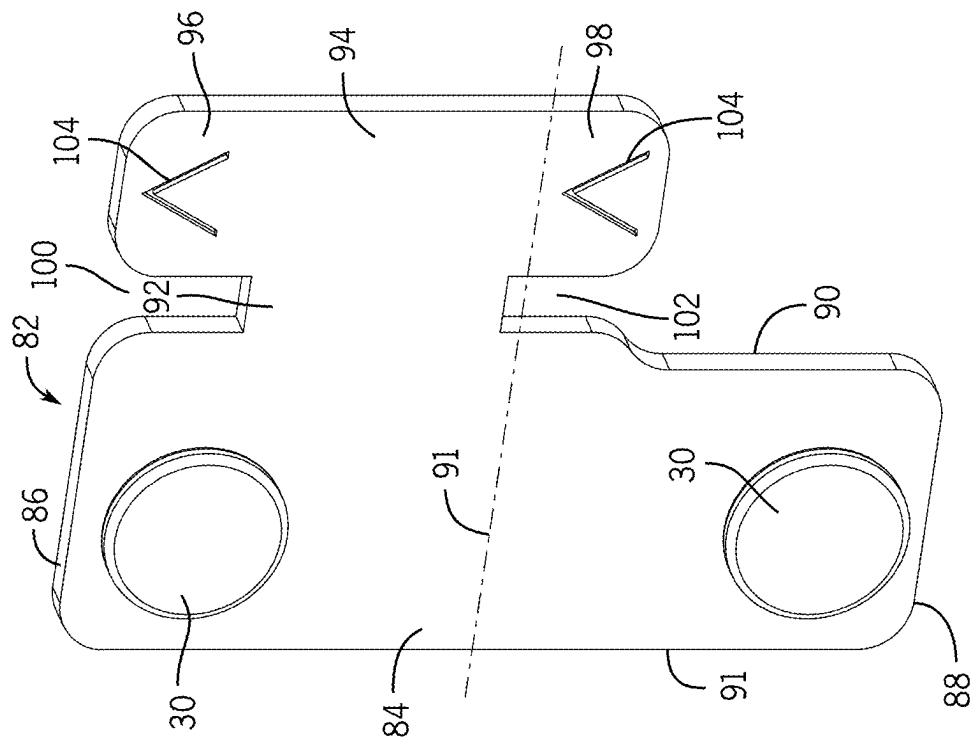


FIG. 16

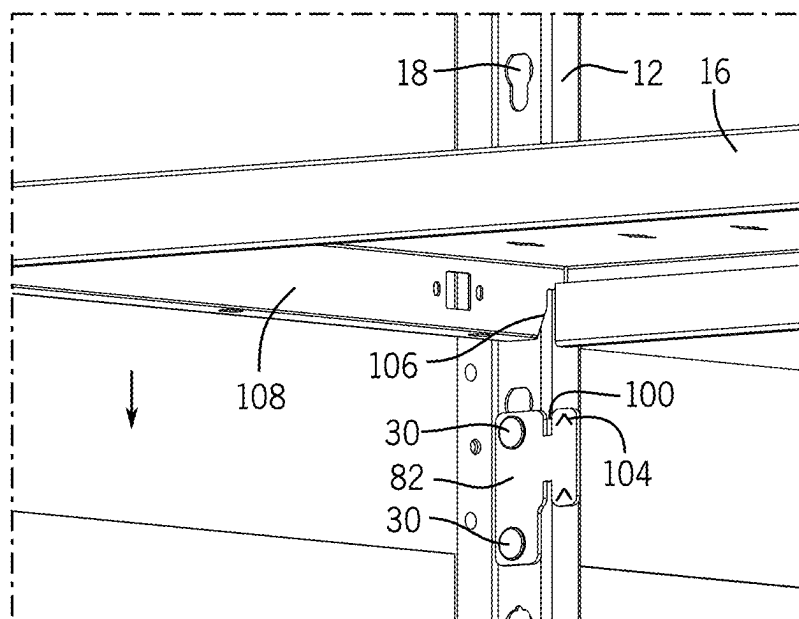


FIG. 17

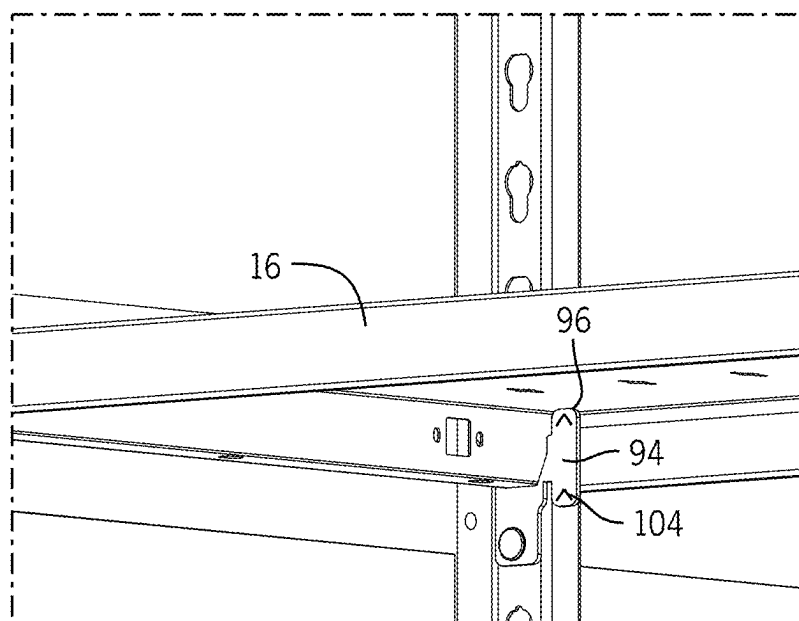


FIG. 18

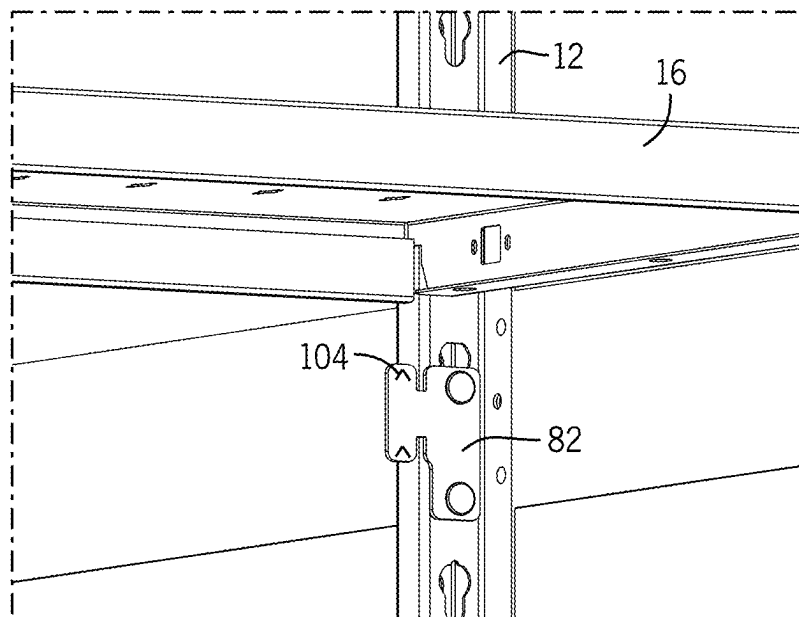


FIG. 19

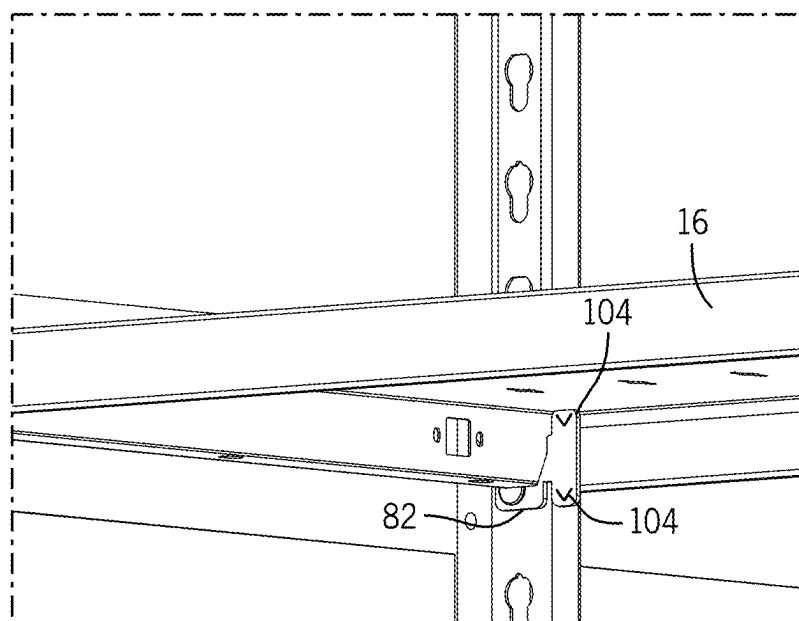


FIG. 20

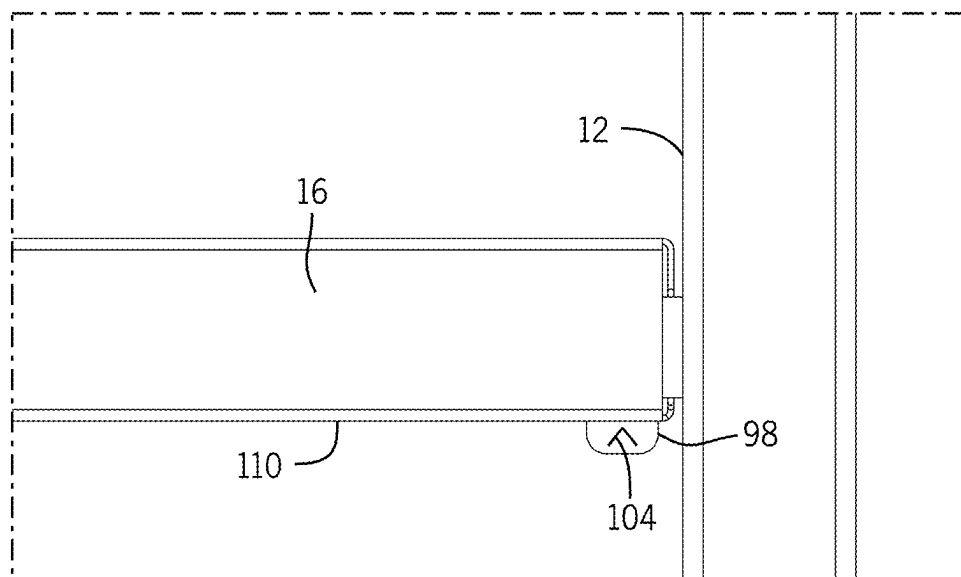


FIG. 21

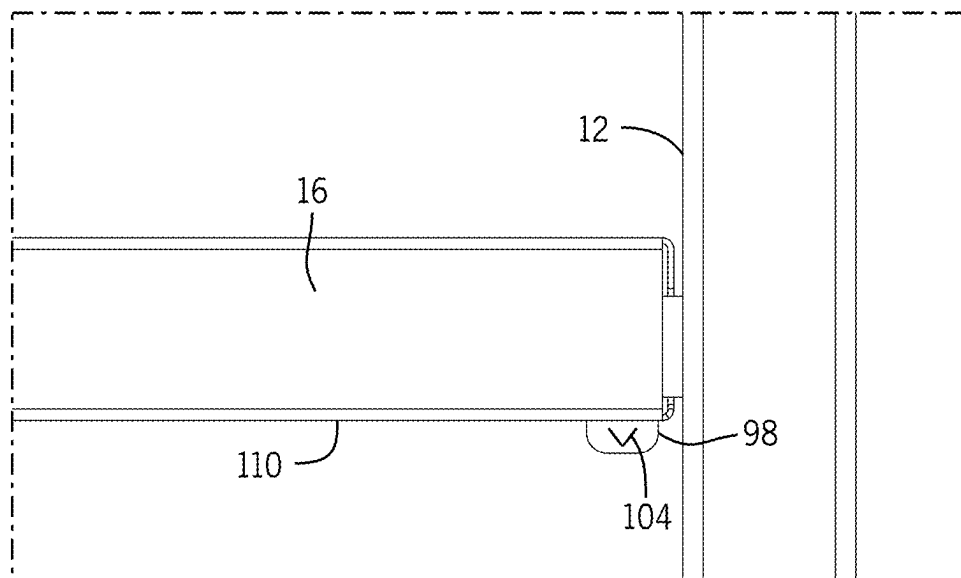


FIG. 22

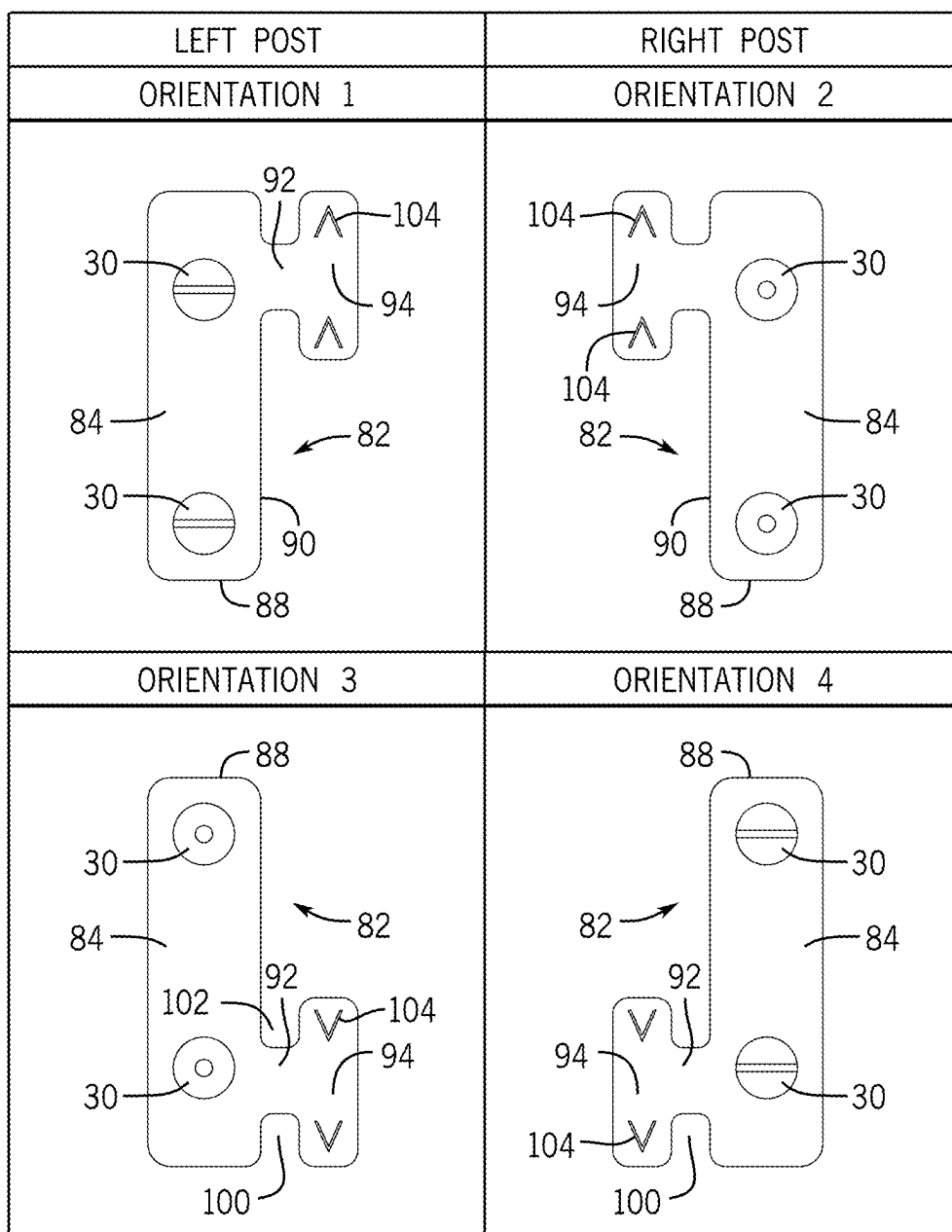


FIG. 23

UNIVERSAL SHELF CLIP FOR STORAGE SYSTEM

CROSS REFERENCE TO RELATED APPLICATION

[0001] The present application is based on and claims priority to U.S. Provisional Patent Application Ser. No. 63/559,454 filed on Feb. 29, 2024, the disclosure of which is incorporated herein by reference.

BACKGROUND

[0002] The present disclosure generally relates to a storage system that includes multiple shelves that are supported between vertical posts. More specifically, the present disclosure relates to a storage system where at least some of the storage shelves can be removably attached to the vertical support posts with shelf clips that can be oriented in at least two different manners to accomplish multiple shelf spacings.

[0003] Current shelf clips on the market are limited to being installed in a single orientation. The single orientation shelf clips provide for fixed spacing between shelves that is dictated by the spacing of cutouts formed in the vertical support posts. While those existing shelf clips may be sufficient if the user is satisfied with the shelf spacing already set by the cutouts in the support posts, problems arise if different or more flexible spacing intervals are desired. It would be impossible to achieve variable configurations outside the prescribed positioning with those rigid, single-orientation clips. For example, a traditional vertical support post that has clip locating cutouts spaced at 1.5" intervals currently limits shelf spacing to integer multiples of the 1.5" spacing (i.e. 3.0" 4.5", 6.0", etc.).

[0004] The asymmetric nature of the various embodiments of the shelf clips of the present disclosure includes a support arm structure that, when combined with slidably attached locating pins, allows for additional intervals to be achieved without the need for shelf clips of different designs. The universality of this shelf connection means there is one part, eliminating the need for multiple parts to set the shelf height at various shelf spacings and eliminates lateralized (i.e. handed) components.

SUMMARY

[0005] The present disclosure generally relates to a storage system that includes multiple shelves that are supported between vertical posts. More specifically, the present disclosure relates to a storage system where at least some of the storage shelves can be removably attached to the vertical support posts with asymmetrical shelf clips that can be oriented in at least two different manners to accomplish multiple shelf spacings.

[0006] The storage system of the present disclosure includes a series of vertical posts that are spaced from each other and are configured to create a support frame that supports a plurality of shelves. Each of the vertical posts includes a series of space cutouts that are designed to receive locking pins of an item to be mounted and supported on the vertical post. The storage system includes a plurality of asymmetrical shelf clips that are designed to be supported on one of the vertical posts and engage one corner or edge of a shelf to support the shelf at the location of the shelf clip.

[0007] Each of the shelf clips includes a clip plate having a first and a second face surface. The clip plate includes

spaced side edges and a support arm extends laterally from one of the side edges of the clip plate. The support arm is configured to engage and support a portion of one of the shelves of the storage system. The support arm is offset from the center line of the clip plate such that the shelf clip can be used in two different orientations in which the support arm is either above or below the center line of the clip plate. The two different orientations of the shelf clip allows for different spacing between shelves supported by the shelf clips.

[0008] In addition to the two different orientations that position the support arm above or below the center line, the shelf clip includes one or more locating pins that can be adjusted to extend from either the first face surface or the second face surface of the clip plate. In this manner, the clip plate can be installed on both right and left vertical posts in each of the first and second orientations. The locking pin includes a pair of head portions located on opposite sides of the clip plate and joined by a center portion that extends through the clip plate. The locking pin can be adjusted such that it extends from either side of the clip plate.

[0009] In accordance with one exemplary embodiment, each side of the shelf clip includes a directional indicator. The directional indicator provides a visual indication of the orientation of the shelf clip both before the shelf clip is installed and after the shelf clip has been installed. The directional indicator thus allows an installer the ability to determine the orientation of the shelf clip to ensure that all of the shelf clips are installed on the vertical posts in the correct and desired orientation.

[0010] The present disclosure is also directed to a shelf clip that can be used with a storage system including a plurality of vertical posts that each include a series of spaced cutouts. The shelf clips are each configured to support a shelf along the height of the vertical posts. Each of the shelf clips includes a clip plate having a first and a second face surface. The clip plate includes spaced side edges, and a support arm extends laterally from one of the side edges of the clip plate. The support arm is configured to engage and support a portion of one of the shelves of the storage system. The support arm is offset from the center line of the clip plate such that the shelf clip can be used in two different orientations in which the support arm is either above or below the center line of the clip plate. The two different orientations of the shelf clip allows for different spacing between shelves supported by the shelf clips.

[0011] In addition to the two different orientations that position the support arm above or below the center line, the shelf clip includes one or more locating pins that can be adjusted to extend from either the first face surface or the second face surface of the clip plate. In this manner, the clip plate can be installed on both right and left vertical posts in each of the first and second orientations. The locking pin includes a pair of head portions located on opposite sides of the clip plate and joined by a center portion that extends through the clip plate. The locking pin can be adjusted such that it extends from either side of the clip plate.

[0012] In accordance with one exemplary embodiment, each side of the shelf clip includes a directional indicator. The directional indicator provides a visual indication of the orientation of the shelf clip both before the shelf clip is installed and after the shelf clip has been installed. The directional indicator thus allows an installer the ability to determine the orientation of the shelf clip to ensure that all

of the shelf clips are installed on the vertical posts in the correct and desired orientation.

[0013] In accordance with yet another exemplary embodiment of the present disclosure, a shelf clip that can be used with a storage system including a plurality of vertical posts that each include a series of spaced cutouts. The shelf clips are each configured to support a shelf along the height of the vertical posts. Each of the shelf clips includes a clip plate extending between a first end and a second end and having a first and a second face surface. The clip plate includes spaced side edges, and a support arm extends laterally from a first side edges of the clip plate. The support arm is configured to engage and support a portion of one of the shelves of the storage system. The support arm is offset from the center line of the clip plate such that the shelf clip can be used in two different orientations in which the support arm is either above or below the center line of the clip plate. The two different orientations of the shelf clip allows for different spacing between shelves supported by the shelf clips.

[0014] The shelf clip further includes a locking tab that is joined to the support arm and is spaced from the first side edge of the clip plate by first and second receiving notches. The receiving notches are each positioned and sized to receive a portion of one of the shelves when the shelf is supported by the shelf clip.

[0015] The shelf clip is designed such that a directional indicator is included on the locking tab on both of the first and second face surfaces of the locking tab. The directional indicator provides a visual indication of the orientation of the shelf clip both before the shelf clip is installed and after the shelf clip has been installed. The directional indicator thus allows an installer the ability to determine the orientation of the shelf clip to insure that all of the shelf clips are installed on the vertical posts in the correct and desired orientation.

[0016] In addition to the two different orientations that position the support arm above or below the center line, the shelf clip includes one or more locating pins that can be adjusted to extend from either the first face surface or the second face surface of the clip plate. In this manner, the clip plate can be installed on both right and left vertical posts in each of the first and second orientations. The locking pin includes a pair of head portions located on opposite sides of the clip plate and joined by a center portion that extends through the clip plate. The locking pin can be adjusted such that it extends from either side of the clip plate.

[0017] Various other features, objects and advantages of the invention will be made apparent from the following description taken together with the drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] The drawings illustrate the best mode presently contemplated of carrying out the disclosure. In the drawings:

[0019] FIG. 1 is a perspective view of a storage system of the present disclosure that includes reversible shelf clips;

[0020] FIG. 2 is a section view of one of the reversible shelf clips used to support shelves of the storage system of the present disclosure;

[0021] FIG. 3 includes views of the shelf clip in two orientations;

[0022] FIG. 4 is a side view of possible locations of shelves based on the use of the shelf clips in two orientations;

[0023] FIG. 5 is a side view of possible locations of shelves based on the use of the shelf clips in only one orientation;

[0024] FIG. 6 is a side view a second contemplated embodiment of the shelf clip;

[0025] FIG. 7 is a view of the second embodiment of the shelf clip attached to one of the vertical posts; and

[0026] FIG. 8 is a side view of the second embodiment of the shelf clip

[0027] FIG. 9 is a view of the shelf clip in the two orientations and as mounted to left and

[0028] right vertical posts;

[0029] FIG. 10 is a perspective view of two alternate embodiments of a shelf clip of the present disclosure;

[0030] FIG. 11 is a view of a first orientation of the alternate embodiment of the shelf clip;

[0031] FIG. 12 is a view of a second orientation of the shelf clip of FIG. 11;

[0032] FIG. 13 is a front view of the two alternate embodiments of FIG. 10 as mounted to one of the vertical posts;

[0033] FIG. 14 is a front view of the shelf clip of two embodiment mounted to one of the vertical posts;

[0034] FIG. 15 is a perspective view of a fifth contemplated embodiment of the shelf clip of the present disclosure in a first orientation;

[0035] FIG. 16 is a perspective view of the fifth contemplated embodiment of the shelf clip of the present disclosure in a second orientation;

[0036] FIG. 17 is a perspective view of the shelf clip of FIG. 15 installed on one of the vertical posts with a shelf in a position to be installed;

[0037] FIG. 18 is a perspective view showing the shelf installed onto the shelf clip shown in FIG. 17;

[0038] FIG. 19 is a perspective view of the shelf clip of FIG. 17 installed on another of the vertical posts;

[0039] FIG. 20 is a perspective view showing the shelf clip of FIGS. 15 and 16 installed on a vertical post in the second orientation;

[0040] FIG. 21 is a side view of the installed shelf supported on the shelf clip in the first orientation of FIG. 15;

[0041] FIG. 22 is a side view of the installed shelf supported on the shelf clip in the second orientation of FIG. 16; and

[0042] FIG. 23 is a view of the shelf clip of FIGS. 15 and 16 in the two orientations and as mounted to left and right vertical posts.

DETAILED DESCRIPTION

[0043] FIG. 1 illustrates a storage system 10 constructed in accordance with a first embodiment of the present disclosure. In the embodiment shown in FIG. 1, the storage system 10 includes a series of vertical posts 12 that are each designed to support one of four individual corners 14 of one or more shelves 16. In the embodiment shown in FIG. 1, the storage system 10 includes two shelves 16 that are spaced from each other along the vertical length of the vertical posts 12. However, the storage system 10 can include a greater number of shelves 16 spaced along the vertical height of the vertical posts 12.

[0044] In the embodiment shown in FIG. 1, the vertical post 12 includes a series of openings or cutouts 18 that are spaced along the entire height of the vertical post 12. Each of the cutouts 18 in the exemplary embodiment illustrated are key-hole openings that have an upper receiving portion

20 and a reduced size retaining portion 22. In the embodiment shown, the vertical support post 12 has cutouts 18 that are spaced at regular intervals, such as but not limited to 1.5", along the entire height of the vertical post 12. The spacing between the cutouts 18 defines the specific locations at which the shelves 16 can be located along the overall height of the vertical posts 12.

[0045] In accordance with the present disclosure, the storage system 10 includes a series of asymmetrical shelf clips 24 that can be used to support the shelves 16. The asymmetrical shelf clips 24 are supported at desired locations along the vertical posts 12 in either a first orientation as shown with the upper shelf 16 or in a second orientation that is shown supporting the lower shelf 16 in FIG. 1. As will be described in greater detail below, selecting the first orientation or the second orientation for the shelf clip 24 will allow a user to change the spacing between the shelves 16 while utilizing the same shelf clips 24.

[0046] FIG. 2 is a side view illustrating the configuration of one of the shelf clips 24. The shelf clip 24 includes a clip plate 26 that is formed from a metallic material and has a first face surface 38 and a second face surface 40 spaced by the thickness of the plate of metallic material. The clip plate 26 extends between a first end 28 and a second end 29. The shelf clip 24 further includes a pair of locating pins 30 that are each mounted within openings 32 formed within the clip plate 26. The center line of each of the openings 32 are spaced the same distance as the distance between the center line of the cutouts 18 formed in the vertical post 12. In this manner, the locating pins 30 are spaced from each other by the same distance as the distance between two of the cutouts 18 formed in the vertical post 12. The spacing between the locating pins 30 allows the locating pins to be received and retained within two of the cutouts 18 formed in the vertical posts 12. Although a pair of locating pins 30 are shown in the embodiment illustrated, the shelf clip 24 could be modified to include either a single locating pin or three or more locating pins. The embodiment including a pair of locating pins 30 has been found to be the most optimal due to material costs, size considerations and strength considerations but is not the only possible embodiment.

[0047] Each of the locating pins 30 includes a pair of head portions 34 that are spaced from each other by a reduced diameter center portion 36. Each of the locating pins 30 is movable relative to the clip plate 26 such that the locating pin 30 can protrude from either of the first face surface 38 or the second face surface 40. The ability of the locating pins 30 to move relative to the clip plate 26 allows the shelf clip 24 to be used in the first and second orientations. In the embodiment shown in FIG. 2, each of the locating pins 30 protrude past the first face surface 38 such that the first face surface 38 comes into contact with the vertical post 12.

[0048] Referring back to FIG. 1, each of the shelf clips 24 includes a support arm 42 that extends laterally from the first side surface 43 of the clip plate 26. The support arm 42 is offset from a center line of the clip plate 26 such that the shelf clip 24 is asymmetrical. The first side surface 43 is located opposite a second side surface 45 that defines the width of the clip plate 26. The lateral support arm 42 extends past the outer edge of the vertical post 12 and is used to support one of the shelves 16. In the first orientation shown in FIG. 1, the support arm 42 is part of the first end 28 and is located above the second end 29.

[0049] In the second orientation of the shelf clip 24 shown in the lower portion of FIG. 1, the shelf clip 24 is flipped vertically such that the support arm 42 is located below the second end 29 and below the center line of the clip plate 26. In each of the two orientations shown, the support arm 42 extends in the same lateral direction relative to the vertical post 12 such that the support arm 42 is able to support one of the two shelves 16. In the side view shown in FIG. 2, the support arm 42 contacts and supports the top wall 44 of the shelf 16.

[0050] FIG. 3 illustrates the two different orientations of the shelf clip 24, where the shelf clip 24 is mounted to the same vertical post, as previously described. In the first orientation, the support arm 42 is located above the second end 29 and above the center line 31 while in the second orientation, the support arm 42 is located below the second end 29 and below the center line 31. In each orientation, the locating pins 30 have the same center line spacing such that the locating pins 30 can be received and retained within the cutouts formed in the vertical posts. Since the support arm 42 is offset from the center line 31 of the clip plate 26, the two orientations shown allow for the different spacing of the supported shelf. FIG. 9 illustrates the two orientations of the shelf clip 24 mounted to either a left support post or to a right support post. The configuration of the shelf clip 24 allows the shelf clip 24 to be oriented such that either of the two face surfaces 38 or 40 to be in contact with the outer wall of the vertical support post and the support arm 42 to be above or below the second end 29. The sliding locating pins 30 allow for this reversible orientation, as described above.

[0051] Referring now to FIG. 4, there is shown one possible use of the two different orientations of the asymmetrical shelf clips 24 to vary the spacing between the individual shelves 16. In the embodiment shown in FIG. 4, the upper shelf 16a is supported by the shelf clip 24 in the first orientation in which the support arm 42 is located above the second end 29. The next vertically lower shelf 16b is supported by the shelf clip 24 in the second, reversed orientation in which the support arm 42 is located below the second end 29. The third shelf clip 24 supporting the lower shelf 16c is again oriented in the first orientation in which the support arm 42 is located above the second end 29. As can be understood by the exemplary spacing illustrated, the reversible shelf clips 24 allow for a variety of different spacing between the individual shelves 16a-16c.

[0052] FIG. 5 is an illustration in which all of the shelf clips 24 are connected to the vertical post 12 in the first orientation. In this manner, the shelf spacing occurs at intervals of 1.5". As can be understood in a comparison of FIGS. 4 and 5, changing the orientation of the shelf clip 24 while utilizing the same two cutouts 18 allows for the spacing between two shelves 16 to be modified as compared to a shelf clip 24 having only a single orientation.

[0053] FIGS. 6-8 illustrates a first alternate, contemplated embodiment of the asymmetrical shelf clip 50 of the present disclosure. In the alternate embodiment shown in FIG. 6, the shelf clip 50 is formed having a similar clip plate 52 that extends between a first end 54 and a second end 56. The shelf clip 50 includes a pair of similar locating pins 30 that have a center line spacing that corresponds to the center line spacing between the cutouts 18 formed in the vertical post. However, unlike the first embodiment previously described the support arm 58 is positioned at a location between the first end 54 and the second end 56. Like the first embodiment

previously described, the support arm 58 is offset from the center line 51 such that the shelf clip 50 is asymmetric. In the embodiment shown, the spacing between the first locating pin 30 and the support surface 59 of the support arm 58 is 0.75" while the spacing between the second locating pin 30 and support surface 61 the support arm 58 is only 0.25". Thus, when the shelf clip 50 is transitioned from the first orientation shown in FIG. 6 to a second orientation, the location of the support arm 58 changes by the difference between the spacing between the two locating pins. FIGS. 7 and 8 show the mounting of the shelf clip 50 to the vertical post in a first orientation in which the support arm 58 is below the first end 54. It should be understood that the shelf clip 50 could be reversed into a second orientation in which the support arm 58 would be above the first end 54. These two orientations are similar to the two orientations of the first embodiment of the shelf clip 24 shown in FIG. 3.

[0054] FIG. 10 illustrates two other contemplated embodiments of a shelf clip 60 and a shelf clip 62. Shelf clip 60 includes a pair of support arms 64 and 66 that are located on opposite ends of the clip plate 68. The clip plate 68 includes a locating tab 70 that is recessed from the clip plate 68 and configured to be received in one of the cutouts 18 formed in the vertical post 12. The spacing between the support arm 64 and the locating tab 70 is different than the spacing between the support arm 66 and the locating tab 70 such that the shelf clip 60 can be used in two different orientations.

[0055] In the shelf clip 62, a pair of locating tabs 70 are recessed from the clip plate 68 and are spaced from each other the same distance as the spacing between the cutouts 18 in the vertical post 12. In the second embodiment, the spacing between the support arm 64 and center point between the pair of the locating tabs 70 is different than the spacing between the support arm 66 and the pair of locating tabs 70 such that the shelf clip 60 can be used in two different orientations.

[0056] FIG. 11 illustrates the shelf clip 60 in the first orientation. As shown, the locating tab 70 is positioned closer to the support arm 64 as compared to the support arm 66. In the second orientation shown in FIG. 12, the shelf clip 60 is flipped such that the support arm 66 is located above the support arm 64.

[0057] FIG. 13 illustrates the two shelf clips 60 and 62 mounted to one of the vertical posts 12 while FIG. 14 illustrates the two shelf clips 24 and 50 mounted to a similar vertical post 12. The comparison of the two Figures illustrates that the support arms 42 and 58 extend past the lateral edge 80 of the vertical post 12 in two embodiments and that the support arms 64 and 66 are located within the outer edges of the vertical post 12. In each embodiment, the orientation of the shelf clip can be changed to allow for multiple shelf spacing distances using a single shelf clip.

[0058] FIGS. 15 and 16 illustrate yet another embodiment of a shelf clip 82 constructed in accordance with another contemplated embodiment of the present disclosure. As with the previous shelf clips described in the preceding figures, the shelf clip 82 includes a clip plate 84 that extends from a first end 86 to a second end 88. The clip plate 84 has a width that is defined by a first side edge 90 and a second side edge 91. The clip plate 84 receives a pair of the locating pins 30 that have the same configuration, shape and spacing as in the previously described embodiments. The center line spacing

between the locating pins 30 corresponds to the center line spacing between two of the cutouts formed in the vertical posts.

[0059] In the embodiment of the shelf clip 82 illustrated, the shelf clip 82 includes a support arm 92 that is offset from the center line 91 and extends laterally from the first side edge 90. The support arm 92 is connected to and joined to a locking tab 94. In the embodiment illustrated, the support arm 92, the locking tab 94 and the clip plate 84 are formed from a single piece of metallic material and can be stamped to create the shape and configuration as illustrated.

[0060] The locking tab 94 includes a first portion 96 and a second portion 98 that extend above and below the support arm 92 when the shelf clip 82 is in the first orientation shown in FIG. 15. The first portion 96 is spaced from the first side edge 90 of the clip plate 84 by a first receiving notch 100 while the second portion 98 is spaced from the side edge 90 of the clip plate 84 by a second receiving notch 102. The dimensions of both of the first and second receiving notches 100, 102 are selected to receive an edge of one of the shelves that are designed to be supported by the vertical posts, in a manner as will be described in greater detail below. The combination of the notches 100, 102 and the locking tab 94 act to secure the shelf in the installed position and to prevent lateral movement of the shelf when installed.

[0061] The shelf clip 82 shown in FIG. 15 includes a pair of directional indicators 104 included on both the first and second face surfaces of the locking tab 94. Only the directional indicators 104 included on the first face surface of the locking tab 94 are shown in FIG. 15 but it should be understood that identical directional indicators 104 are on the opposite second face surface that is not shown. Each of the directional indicators 104 has a shape and configuration that indicates an orientation direction of the shelf clip 82. In the embodiment shown in FIGS. 15 and 16, the directional indicators 104 are arrowheads that point in a defined direction. In the embodiment shown in FIG. 15, the arrow-shaped directional indicators 104 point upward when the shelf clip 82 is in the first orientation. In the second orientation shown in FIG. 16, the arrow-shaped directional indicators 104 point vertically downward. In this manner, a user and/or assembler can determine the orientation of the shelf clip 82 both before and after it is installed on one of the vertical posts. The directional indicators 104 provide an easier way for the person assembling the shelving unit to determine the orientation the shelf clip 82 both before and after the shelf clip has been installed and/or when a shelf is received on the shelf clip 82.

[0062] FIGS. 17 and 18 illustrate the installation of one of the shelves 16 on one of the vertical posts 12 utilizing the shelf clip 82. As indicated previously, each of the vertical posts 12 includes a series of cutouts 18 that are evenly spaced along the overall height of the vertical post 12. In the embodiment shown in FIG. 17, the shelf clip 82 is installed in the orientation as shown in FIG. 15. In this orientation, each of the locating pins 30 are received and retained within one of the cutouts 18 such that the locating pins 30 securely hold the shelf clip 82 in the position shown. As can be seen in FIGS. 17 and 18, the directional indicators 104 point vertically upward such that the installer can quickly determine the orientation of the installed clip 82.

[0063] Once a shelf clip 82 is installed on each of the four spaced vertical posts 12 in the first orientation with the directional indicators 104 point upward, the shelf 16 can be

lowered such that a receiving slot **106** formed in the end wall **108** near each of the four corners of the shelf is aligned with the first receiving notch **100**. As the shelf **16** is moved vertically downward as shown by the arrow in FIG. 17, the receiving slot **106** enters into the first receiving notch **100** until the upper end of the receiving slot **106** contacts the top edge surface of the support arm **92**. Once in this position, the first portion **96** formed on the locking tab **94** helps to lock the shelf in place and prevents lateral movement of the shelf. In this locked position, both of the directional indicators **104** are viewable from below the shelf **16** as illustrated.

[0064] FIG. 19 illustrates another one of the shelf clips **82** installed in one of the spaced vertical posts **12**. As illustrated, the directional indicators **104** provide a visual indication that the shelf clip **82** shown in FIG. 19 is installed in the same first orientation as the shelf clip **82** shown in FIG. 17. Thus, shelf clips **82** installed on different vertical posts **12** can be easily inspected to ensure that each of the shelf clips **82** are installed in the proper orientation.

[0065] FIG. 20 illustrates one of the shelf clips **82** installed in the second orientation, such as illustrated in FIG. 16. As shown in FIG. 20, the directional indicators **104** provide a visual indication to the installer as to the orientation of the installed shelf clip **82**. The installer can then install additional shelf clips **82** in the same second orientation to correctly support one of the shelves.

[0066] FIGS. 21 and 22 provide an end view of one of the shelves **16** after it is installed on one of the vertical posts **12**. As can be seen in both of the FIGS. 21 and 22, the directional indicator **104** is visible below the lower edge **110** of the shelf after the shelf **16** has been installed. In this manner, the installer can verify the correct installation of the shelves and the shelf clips **82** after assembly. In the embodiment shown in FIG. 21, when the shelf clip is in the first orientation, the second portion **98** of the shelf clip **82** is visible below the lower edge **112**. When the shelf clip **82** is in the second orientation shown in FIG. 22, the first portion **96** is visible. As can be understood in the comparison of FIGS. 21 and 22, the directional indicator **104** provides an indication as to the orientation of the shelf clip **82**.

[0067] FIG. 23 provides an illustrative indication of the various orientations of the shelf clip **82** as mounted in both first and second vertical orientations, as well as when mounted to either a left vertical post or a right vertical post. In the illustration shown as orientation 1, the locking tab **94** is located to the right of the clip plate **84** and the support arm **92** is positioned above the center line of the clip plate **84**. In this orientation, each of the directional indicators **104** point upward. When a shelf is to be installed utilizing the shelf clip **82** attached to the left vertical post in the first orientation, another shelf clip **82** is installed on the right post as indicated by orientation 2 of FIG. 23. To create this orientation, the shelf clip **82** is flipped such that the second face surface contacts the vertical post, which results in the locking tab **94** extending to the left of the clip plate **84**. In accordance with the embodiment of the present disclosure, the directional indicators **104** are included on both the first and second face surfaces of the locking tab **94** spaced from the clip plate **84**. Thus, when the shelf clip **82** is rotated from orientation 1 to orientation 2, the directional indicators **104** are still visible and point vertically upward.

[0068] In orientation 3 shown in FIG. 23, the shelf clip **82** is flipped downward such that the locking tab **94** is located below the center line of the clip plate **84**. In this embodi-

ment, the first receiving notch **100** is located below the center line while the second receiving notch **102** faces upward. As can be understood in the comparison between orientation 1 and orientation 3, the support arm **92** is located below the center line in orientation 3 while the support arm **92** is located above the center line in orientation 1. In this manner, the location of a shelf supported by the shelf clip **82** can be adjusted by transitioning the shelf clip **82** from orientation 1 to orientation 3. When the shelf clip **82** is in orientation 3, the directional indicators **104** on the first face surface of the locking tab **94** each point downward, thus signifying the difference between the orientations of the shelf clip **82**.

[0069] Orientation 4 shown in FIG. 23 shows the use of the shelf clip **82** attached to a right post. In this orientation, each of the two directional indicators **104** contained on the second face surface of the locking tab **94** face downward to provide a visual indication of the orientation of the shelf clip **82**.

[0070] In much the same method and concept as shown in FIGS. 4 and 5 discussed previously, the shelf clip **82** can be mounted to one of the vertical posts **12** in either the first orientation or the second orientation to change the spacing between the individual shelves. When the shelf clips **82** are positioned in the first orientation, the spacing between the shelves will be as shown in FIG. 5. If one of the shelf clips **82** is transitioned to the second orientation, such as shown in FIG. 16, the spacing between the individual shelves can be varied. In this manner, the use of the asymmetrical shelf clip **82**, in the same manner as the previous asymmetrical shelf clips described previously, can be used to vary the spacing between individual shelves.

[0071] This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to make and use the invention. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

We claim:

1. A storage system comprising:

- a series of vertical posts, each post having a series of cutouts;
 - a plurality of shelf clips, each shelf clip having a clip plate, a support arm extending laterally from the clip plate, and one or more locating pins extending from the clip plate and insertable into the cutouts of the vertical posts; and
 - a plurality of shelves each supported along the vertical posts by a plurality of the reversible shelf clips,
- wherein each of the shelf clips can be installed along one of the vertical posts in at least two different orientations to achieve variable spacing between the plurality of shelves.

2. The storage system of claim 1 wherein each of the shelf clips includes at least two locating pins that are spaced from each other at a distance that corresponds to spacing between the series of cutouts on the vertical posts.

3. The storage system of claim 2, wherein the cutouts in the vertical posts are key-hole openings having an upper receiving portion and a reduced size retaining portion.

4. The storage system of claim 1, wherein the shelf clips are asymmetrical and the support arm is offset from a center line of the clip plate.

5. The storage system of claim 1, wherein the shelves are spaced from each other along the vertical length of the posts and the spacing between the shelves can be changed by selecting the orientation of the shelf clips.

6. The storage system of claim 1, wherein the shelf clips are reversible and can be installed on the vertical posts in a first orientation or a second orientation.

7. The storage system of claim 6 wherein each of the shelf clips includes a directional indicator to indicate the orientation of the shelf clip when the shelf clip is installed on the vertical post.

8. A shelf clip for use in a storage system including a plurality of vertical posts each including a series of spaced cutouts and one or more shelves supported between the vertical posts, the shelf clip comprising:

a clip plate;

a support arm extending laterally from the clip plate; and one or more locating pins extending from the clip plate, wherein each of the shelf clips can be oriented in at a first orientation and a second orientation to achieve variable spacing between the shelves along the vertical posts.

9. The shelf clip of claim 8, wherein the support arm extends from the clip plate at a location offset from a center line of the clip plate, wherein when the shelf clip is installed in the first orientation, the support arm is vertically above the center line and when the shelf clip is installed in the second orientation the support arm is vertically below the center line.

10. The shelf clip of claim 8, wherein at least two locating pins extend from the clip plate and are spaced from each other at a distance that corresponds to a distance between the series of spaced cutouts in the vertical post.

11. The shelf clip of 10 wherein each of the at least two locating pins include a pair of head portions joined to each other by a center portion that extends through the clip plate.

12. The shelf clip of claim 11 wherein each of the at least two locating pins are slidable within the clip plate to vary the spacing between each of the pair of head portions and the clip plate.

13. The shelf clip of claim 8 wherein the shelf clip can be installed along one of the vertical posts such that the support

arm extends in either a first direction or an opposite second direction in either the first orientation or the second orientation of the shelf clip.

14. The shelf clip of claim 8 wherein the shelf clip includes a directional indicator to indicate the orientation of the shelf clip when the shelf clip is installed on the vertical post.

15. A shelf clip for use in a storage system including a plurality of vertical posts each including a series of cutouts spaced at a distance and one or more shelves supported between the vertical posts, the shelf clip comprising:

a clip plate extending between a first end and a second end and including a first side edge and a second side edge; a support arm extending laterally from the first side edge clip plate;

a locking tab joined to the support arm and being spaced from the first side edge by a first receiving notch and a second receiving notch; and

at least two locating pins extending from the clip plate and spaced from each other at the distance of the spacing of the series of cutouts, wherein each of the shelf clips can be oriented in at a first orientation and a second orientation to achieve variable spacing between the shelves along the vertical posts.

16. The shelf clip of claim 15 wherein the support arm extends from the clip plate at a location offset from a center line of the clip plate, wherein when the shelf clip is installed in the first orientation, the support arm is vertically above the center line and when the shelf clip is installed in the second orientation the support arm is vertically below the center line.

17. The shelf clip of claim 15 wherein each of the at least two locating pins include a pair of head portions joined to each other by a center portion that extends through the clip plate.

18. The shelf clip of claim 17 wherein each of the at least two locating pins are slidable within the clip plate to vary the spacing between each of the pair of head portions and the clip plate.

19. The shelf clip of claim 18 wherein the shelf clip can be installed along one of the vertical posts such that the support arm extends in either a first direction or an opposite second direction in either the first orientation or the second orientation of the shelf clip.

20. The shelf clip of claim 15 wherein each of the shelf clips includes a directional indicator to indicate the orientation of the shelf clip when the shelf clip is installed on the vertical post.

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